



EAST WEST UNIVERSITY
Department of Mathematical and Physical Sciences (MPS)
Semester: **Summer-2026**

Course Outline: MAT205

Course Title : Linear Algebra and Complex Variables
Course Code : MAT205
Pre-Requisite : MAT102 and MAT104
Credit Hours : 3
Course Instructor : Prof. Md. Ashraful Alam (Alam)
Office Room : FUB-1004, Dept. of MPS
Cell : 01728201266 (Whatsapp)
Email Address : ashraful.alam@ewubd.edu

Google

Classroom Code :

**Class Schedule
and Office Hour**

Day	Course No.	Section	Class Time	Room No	Office Hour
MW	MAT 205	7	1.30 pm – 3.00 pm	FUB-504	3.10 pm - 4.40 pm
	MAT 205	8	4.50 pm – 6.20 pm	FUB-802	Dept. of MPS Room No. FUB-1004

Course Goal:

For the students of any branch of science, knowledge of Mathematics is essential. Linear Algebra develops the basic concept on matrix, determinant, vector space, linear transformation and some other important topics, which will help the students to develop the analytical ability. Theories of Complex Variables are very much important for all branches of Science and Engineering. This will help in developing the theories of Laplace Transformation, Fourier series and Fourier Transformation.

Learning Outcomes:

On successful completion of this course students will be able to

- solve systems of linear equations by using Gaussian elimination and Gauss-Jordan elimination method;
- be able to apply the basic techniques of matrix algebra, including finding the inverse of an invertible matrix using adjoint and row canonical form;

- understand the basic ideas of vector algebra: linear dependence and independence and spanning;
- be able to find the eigenvalues and eigenvectors of a square matrix using the characteristic polynomial and will know how to diagonalize a matrix when this is possible;
- understand the significance of continuity and differentiability for complex functions and be familiar with the Cauchy-Riemann equations;
- compute the Taylor and Laurent expansions of simple functions, determining the nature of the singularities and calculating residues using the Cauchy Residue Theorem;
- state and prove versions of Cauchy's theorem and its consequences including Cauchy's integral formula, the power series representation for analytic functions and the Fundamental Theorem of Algebra.

Course Contents : 1. Linear Algebra.
2. Complex Variables.

Description : The following topics will be covered throughout the semester.

Linear Algebra: Matrices and operations with matrices, Determinants, Systems of linear equations; Vector spaces; Linear independence; Basis and dimension; Linear transformations; Eigenvalues and eigenvectors; Inner Product Space, Diagonalization of matrices; LU-decomposition; Applications.

Complex Variables: Functions of a complex variable and their derivatives; Analytic function; Singularities; Cauchy integral theorem and formula; Power series and Laurent expansions; Calculus of residues and contour integrals; Applications.

Text books : 1. Elementary Linear Algebra-Applications Version, Howard Anton, Chris Rorres, 11th Edition, Wiley-India.
2. Complex Variables, Murray R Spiegel, Seymour Lipschutz, Schaum's Outline Series, Second Edition, McGraw-Hill.

Reference books : (a) Linear Algebra, 5th Edition, Seymour Lipschutz and Marc Lipson, Schaum's Series, McGraw-Hill.
(b) Elementary Linear Algebra, 6th Edition, RON LARSON and DAVID C. FALVO, Houghton Mifflin Harcourt Publishing Company, *Boston, New York*.
(c) Complex Variable by M.L. Khanna.

Term Examinations:

Examinations	Day	Date
Midterm	Monday	06 July 2026
Final	Monday	24 August 2026

Score Distribution:

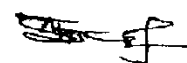
Quiz	Assignment	Class Attendance	Class Performance	Midterm	Final
20%	10%	5%	5%	30%	30%

Grading Policy:

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
less than 40%	F	0.00

Special Instructions:

- Students are not allowed to enter into the classroom after 5 minutes of starting time.
- No make-up quizzes and assignments will be held.
- Students are requested to switch off their mobile during the class hour.
- ***There is zero tolerance for cheating at EWU. Students caught with cheat sheet in their possession, whether used or not used, &/or copying from cheat sheets, writings on the palm of hand, back of calculators, chairs or nearby walls, etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.***



(Md. Ashraful Alam)

Date: May 14, 2026